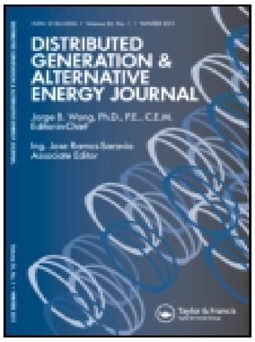

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Optimal Hybrid PV-Battery-Diese Generator Energy System for the Oil Producing Communities in Niger-Delta, Nigeria: A Case Study

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ABSTRACT

Optimal configuration and design of a hybrid Photovoltaic (PV)-Battery-Diesel-Generator energy system has been proposed to power households in Omavovwe community in the Niger-Delta region of Nigeria. The configuration of the optimal hybrid system is selected based on the Hybrid Optimisation Model for Electric Renewable (HOMER) top-ranked system configuration, according to the net present cost. An optimal system design delivers the best components alongside appropriate operating strategies to provide the most efficient, reliable cost-effective system possible. The system investigated reduces CO_2 emissions by 25.72%/year. This will reduce costs imposed on CO_2 emissions by future environmental legislation. The system has a better potential for providing the energy needs of the households considered in this article compared to a relatively high PV penetration (about 74%) as capital costs are reduced by 45.6%. It is observed that initiatives by the federal government of Nigeria (FGN) and the oil producing companies for the utilisation of PV energy in the oil producing communities would realize improved hybrid energy system economic indices.

Keywords: Hybrid PV-Battery-Diesel generator system; optimal configuration; system design; renewable energy.

Nomenclature

AC	Alternating current
BD	Battery-diesel
COE	Cost of energy
DC	Direct current
DG	Diesel generator
FGN	Federal Government of Nigeria
HOMER	Hybrid optimisation model for electric renewable
NPC	Net present cost
NREL	National renewable energy laboratory
PV	Photovoltaic
RES	Renewable energy system
US	United states

INTRODUCTION

Although Nigeria (located in 10:00º N and 8:00°E), constantly referred to as the *Giant of Africa*, is the largest African oil exporter and the fifth among the Organization of Petroleum Exporting Countries (OPEC), over 70% of its population lives in poverty. The geographical area covered by the country is 923,770km^2 of which total land area is 910,770km^2; and the extensive coastline of Nigeria shares an approximate area of 853km^2. The South-South geo-political zone, which is one of the six geo-political zones, comprises solely the Niger-Delta region. The Niger-Delta region covers an area of well over 70,000 km^2 that forms the largest wetland in Africa. It is rich in both renewable and non-renewable natural resources. The Niger-Delta region produces 100% of the over 70% petroleum (oil and gas) revenue contribution of the Federal Government of Nigeria (FGN) revenue [1]. With over 35 million barrels in oil reserves and producing about 2.5 million barrels per day, the Niger-Delta region in the southern Nigeria is a major crude oil exporter to both the US and Western Europe. The petroleum continues to maintain its prominence as the single most important source of the FGN revenue and foreign exchange earner. Despite the apparent abundance of such a coveted and lucrative resource, however, the economic situation in Nigeria is appallingly non-correlative. The non-correlative of the Nigerian economy with its apparent wealth is attributed to the declining of other sectors of the FGN's. For example production of electricity declined by 13.4% between 2002 and 2006 [1].

Adequate electric power production plays a key role in national development; however, there is a gross electric power shortage in Nigeria, which required about 90% of Nigerian businesses to own diesel generators for their daily energy demands, according to The World Bank report [2]. The situation is worrisome for the remote villages, which are located far away from the national utility grid. In particular the oil producing communities of the Niger-Delta region, which are mainly located away from towns, are largely in the rugged terrains (see Figure 1 for the Niger-Delta map). These remote oil producing communities normally, outside the environmental and health concerns [3], lack in roads, running water and ultimately supply of electricity [4], since it is considered impractical to extend the national grid to these dispersed populated areas. Meanwhile, electricity is required for such basic developmental services such as pipe borne water, health care, telecommunications and quality education. More so, the poverty eradication and Universal Basic Education programmes of the FGN, which are targeted on the rural communities, need electricity to be successful. The absence of electricity supply in the oil producing communities has not only left the dwellers of the communities socially backward, but has left their economic potentials untapped. However, majority of the oil producing communities depend on diesel generators and firewood for their daily electricity and cooking demands, respectively. Firewood is used by over 60% of the Niger-Deltans, which dwell predominantly in the oil producing communities. Sourcing firewood for domestic usage, basically for cooking, is attributed, in part, to be cause of erosion from deforestation in the Niger-Delta. The rate of deforestation is about 350,000 hectares/year, which is equivalent to 3.6% of the present area of forests and woodlands; whereas reforestation is only at about 10% of the deforestation rate, according to the Nigerian Inter-Ministerial Committee on Deforestation report [5]. It is worth noting that the average official pump price of diesel in Nigeria is about $1.00/Litre; however, this price increases almost by 50% ($1.50/Litre) in the oil producing communities seemingly due to the rugged terrains and the spurious notion of high economic attachment to the oil producing communities' dwellers. Since absolute majority of the oil producing communities totally depend on diesel generator and firewood for energy supply, it implies that the combined effect of stringent Government's policy against indiscriminate deforestation and global hike in fuel prices would have undesirable impact on the social-economic position of the communities—especially increase the crime rate embroiled Niger-Delta region that is attributed to the decades neglect of the region by the FGN and the oil producing companies [4].

Therefore, Renewable Energy Sources (RES) as alternative or complement energy sources in these oil producing communities would be of great benefits, especially the attendant concerns for global warming as a result of conventional energy conversion systems play important roles in the general acceptance of RES. Since the supply of RES is transient, the search to store the harnessed energy is paramount. Integrating hybrid RES into buildings in the oil producing communities would provide reliable and secured energy supply. One of the most promising options is the utilisation of hybrid PV-battery-diesel generator (PV-BD) energy system, since the Niger-Delta region is within an abundant solar resource (3.2-5.5 kWh/m^2/day). Hence, buildings integrated hybrid PV-BD energy system for oil producing communities is long overdue.

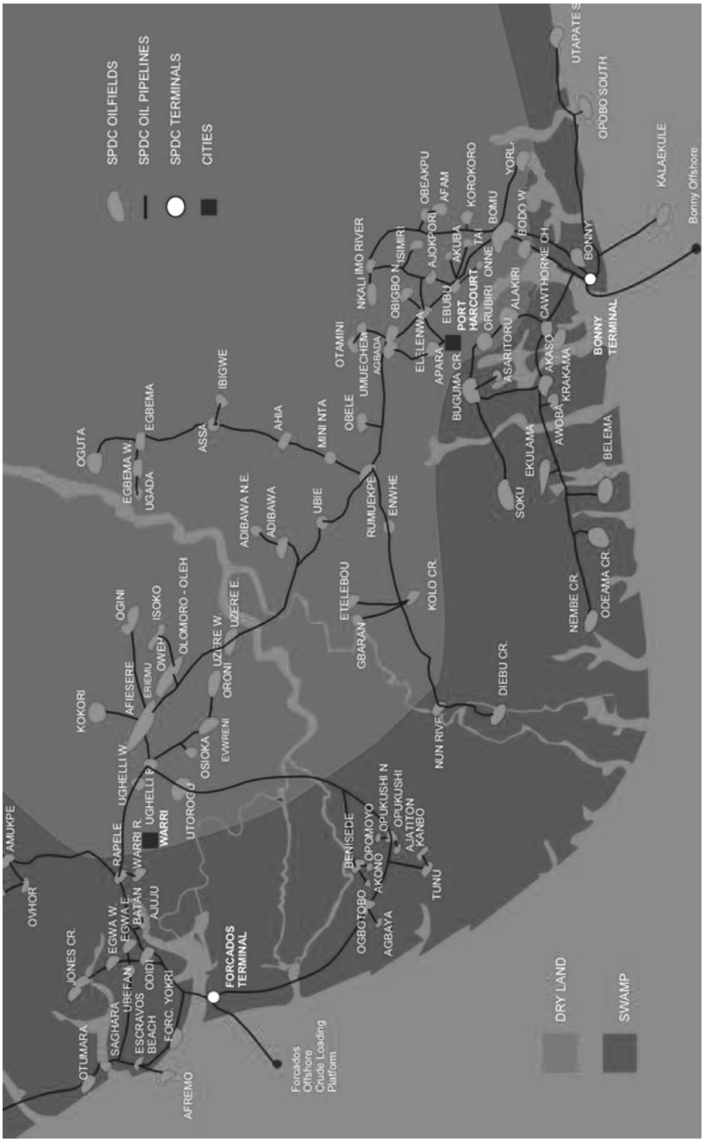


Figure 1: The Niger-Delta showing some of the Nigerians' oil producing communities [6]

Optimisation studies on hybrid PV-BD systems utilisation can be found in [7], [8]. An hourly solar energy series, a model of hybrid PV-DB system and an hourly load profile are normally used in the optimisation of hybrid PV-BD systems. Therefore, this article proposed an optimal hybrid PV-BD energy system for households in an oil producing community in the Niger-Delta region, Omavovwe Community (with geographical coordinates of 5.32°N, 6.46°E) to be specific, on the basis of the energy demand of an average household in the community. The choice of complementing diesel generator with the PV energy is on the fact that: (1) the area is located in abundant solar irradiation zone, 3.2-5.5 kWh/m^2/day, (2) the establishment of 7.5MW joint venture solar panel manufacturing plant in Nigeria by the National Agency for Science and Engineering Infrastructure Development Agency (NASEI), (3) proposed establishment of solar PV maintenance shops across the Niger-Delta region by the Niger Delta Development Commission (NDDC) as a deliberate effort to launch the region into the world of renewable energy technology and (4) ultimately, to reduce the pollutant emissions from the use of diesel generators, as a way of salvaging the over-polluted region due to the activities of oil producing companies.

SYSTEM DESCRIPTION

The hybrid RES comprises solar PV panel, diesel generator, battery pack, charge controller and converter (inverter and rectifier), as depicted by Figure 2. The solar PV panel produces Direct Current (DC) from sunlight, which is stored in the battery pack through the charge controller. The brown colour flow line, in Figure 2, depicts the flow of DC current; whereas the red flow line depicts the flow of Alternating Current (AC). The diesel generator (DG) converts chemical energy (diesel fuel) to electrical energy (AC current) through mechanical energy and supplied directly to the household; however, the DG equally charges the battery pack, when it runs at full power (cycle charging control strategy), via the rectifier. The electrical energy stored in the battery, in the form of chemical energy, is converted into AC power by the converter to meet the electrical load demand of the household. The elements function together to form the entire hybrid PV-BD system capable of supplying electric power to households in the Omavovwe community.

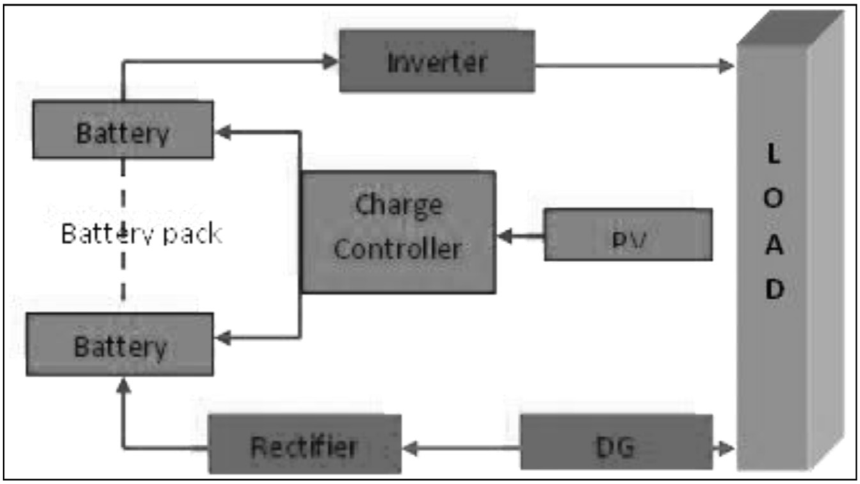


Figure 2: Description of the hybrid PV-BD energy system.

In this configuration, the load can be powered either by the DG or by the PV system through the battery pack-inverter flow line. It is expected that both the PV system and the DG would charge the battery bank. The DG, however, is able to directly supply the facility load. There will be an embedded automatic switch, which would shut down the DG when the facility load demand is lower than the supply of PV-battery system only and vis-à-vis starting the DG when load demand is above the PV-battery supply.

DATA COLLECTION

Case studied/Solar Irradiance

Solar energy is a reliable source of energy that is received in forms of beam and diffuse radiations. The irradiance reaching the earth's surface depends on the cloudiness or clearness of the sky, which in turn depends on the season of the year. However, harnessing the insolation for useful energy is vital to the solar design engineers [9] [10]. Omavovwe community is a typical oil producing community in the Niger-Delta region of Nigeria. The community is located in geographical coordinates of 5.32°N, 6.46°E. The community, shown in Figure 3, is located in an island called *Okpaode*; it has features that bore true representation of oil producing communities found in the Niger-Delta region; e.g. houses,

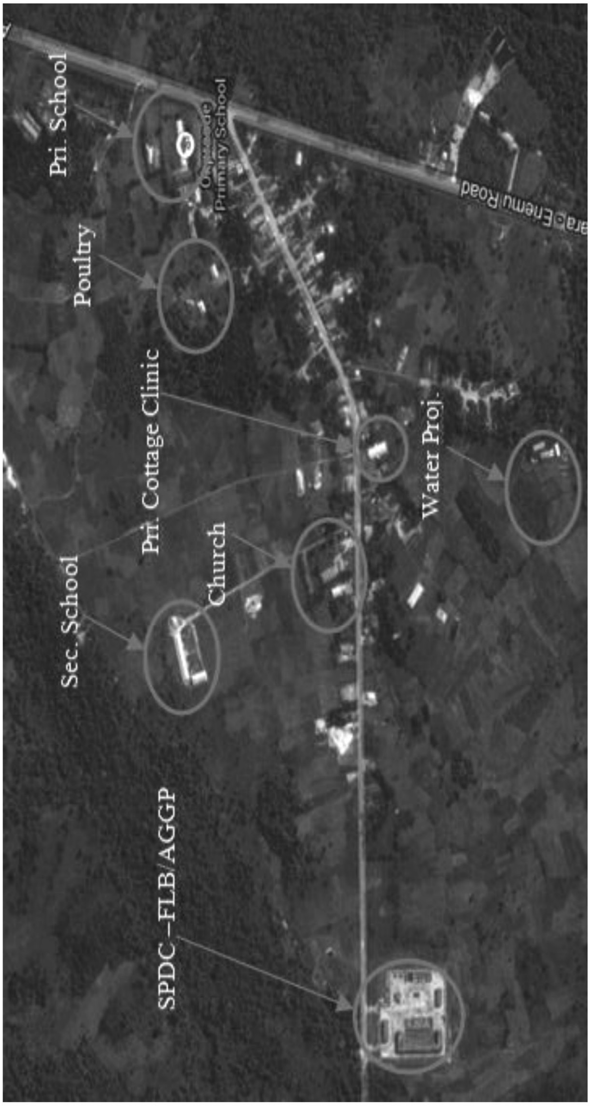


Figure 3: Google satellite image of Omavovwe community

primary school (Pri. School), secondary school (Sec. School), church, village hall, poultry, private cottage clinic (Pri. Cottage Clinic), ongoing solar water project (Water Proj.), Shell Petroleum Development Company (SPDC) field logistic base (FLB) and proposed associated gas gathering plant (AGGS). The Omavovwe community has over a hundred habitable households.

Based on the solar irradiance on horizontal plane of the area [11] and the HOMER software, Figure 4 is obtained for the Omavovwe community location. It can be seen from Figure 4 that more solar irradiance is expected from the month of November to April, whereas from May to October the solar irradiation would be minimal. This observation corresponds to the dry season and the wet season, respectively.

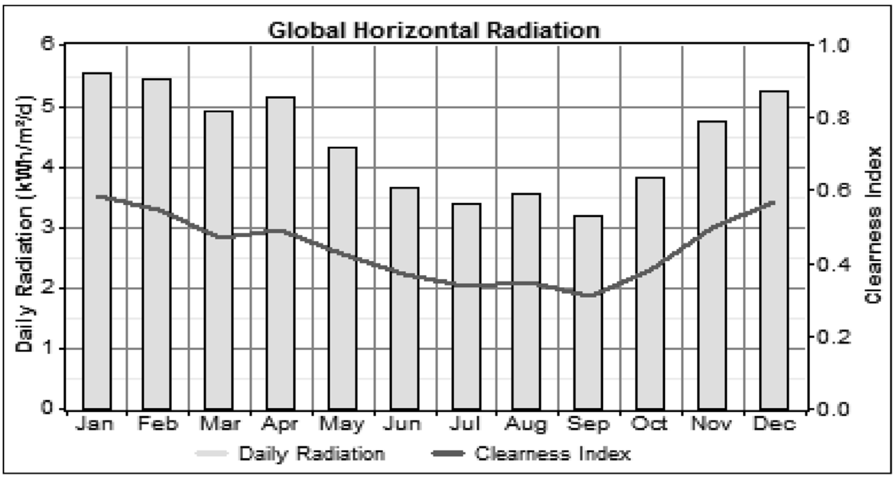


Figure 4: Solar irradiation of the community

Diesel Generator

DGs are mechanical devices that convert chemical energy to electrical energy via mechanical energy. Usually, these generators require high running cost, frequent maintenance and they pollute the environment. The ratings of the DG are determined by the energy demand and the renewable energy penetration, which is normally within the range of 11-25% [12]. The price of diesel fuel, which is used to power diesel generators, in Nigeria is relatively high since it amounts to about US$1.00/Litre; however, this price is increased almost by 50% (to amount to US$1.50/Liter) in the oil producing communities.

Electrical Load Estimation

In this article, the hybrid PV-BD electric power is intended to be localised to each household as against the concentration of the power in the community. This is aimed to reduce maintenance cost as a centralised Hybrid PV-BD is normally accompanied by relatively high maintenance cost [13]. The non-optimised power load profile shown in Figure 5 represents a typical household in the community. The line indicates the total load profile and the bars represent the contributing appliances.

The profile was recorded day-to-day and monitored over time to compute the average daily energy requirement of the household. It should be noted that water pump power is included in the daily energy profile, as an average household has autonomous water supply, necessary because the water project begun in the village over 15 years ago by the then Oil Mineral Producing Area Development Commission (now Niger Delta Development Commission—NDDC) is still far from completion. The load profile (Figure 5) shows that maximum demand occurs from 11am to 2 pm, which corresponds to the simultaneous working hours of the water pump and the refrigerator. The water pump, which operates for about two hours, is used to pump water into overhead water storage, which supplies neighbourhoods, and is used for the whole day. On average, electric power demand is high between 2-10 pm, which corresponds to the time when students and farmers return to their homes; whereas the demand is low between 10 pm-6 am, which corresponds to sleeping time.

The equipment, lightings, and other electrical devices used in a typical household, with their corresponding power ratings, are given in Table 1. The daily energy requirement of the household is obtained to be 12 kWh by the HOMER software – day-to-day and time-step-to-time-step variability are assumed to be 5%.

Figure 6 further shows the seasonal variation of the load profile, which is based on the day-to-day and time-step-to-time-step variability factor. The figure shows the daily high, daily mean and daily low electric power demand. The seasonal variation is due to the two major seasons in Nigeria – Wet Season and Dry Season. The wet season is accompanied with rains in the months of May, June, July and, occasionally, October. The wet season is relatively cold; hence, it is assumed that the use of fan is not necessary. The dry season comes in two conditions – cold and hot. It is assumed that the use of fan is not necessary in the dry-cold season (the Harmattan period), which is, normally, from December to February.

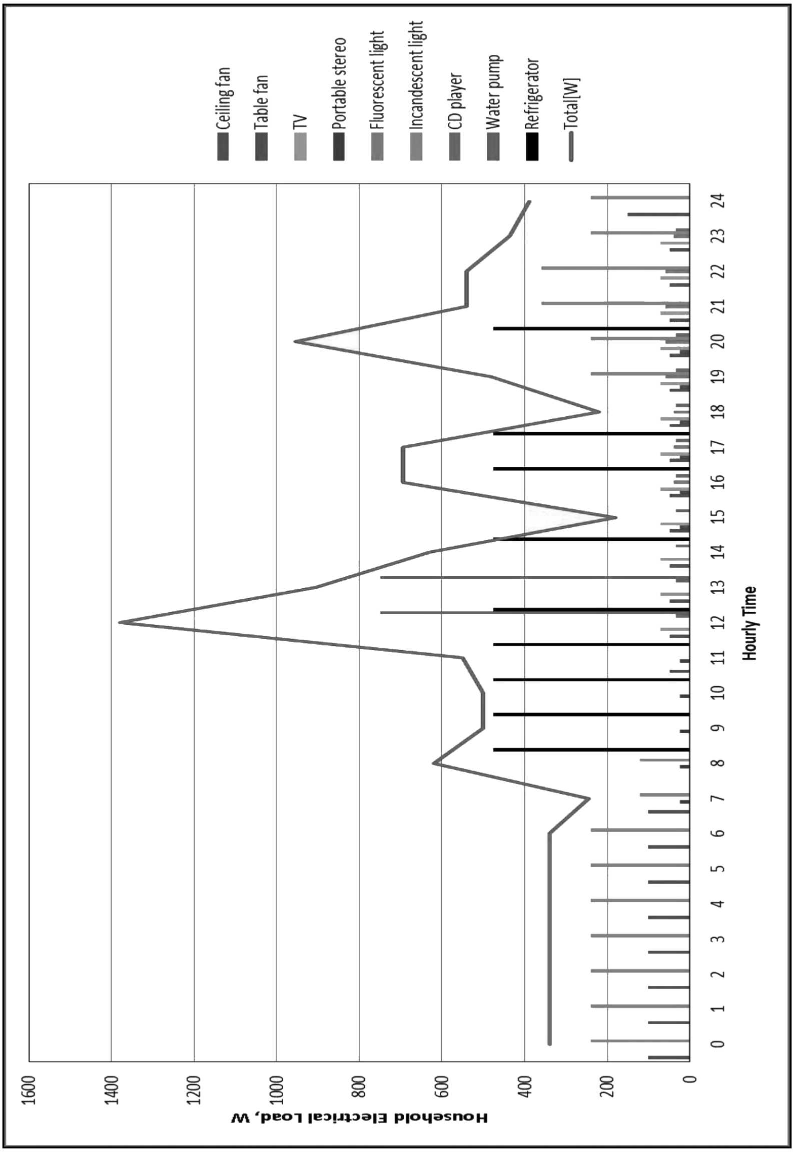


Figure 5: Non-optimised average household electric power profile

Table 1: Power rating of equipment per household

k	Appliance	Units, U_k	Wattage [W], $\dot{W}_k$	Hour Used/day, H_k
1	Ceiling fan	3	50	10.7
2	Table fan	1	25	6
3	TV (19" colour)	1	70	12
4	Portable stereo	1	25	5
5	Fluorescent light	4	20	3.25
6	Incandescent light	8	60	7.5
7	Refrigerator ($0.45307m^3$)	1	475	7
8	CD Player	1	35	10
9	Water pump	1	750	2

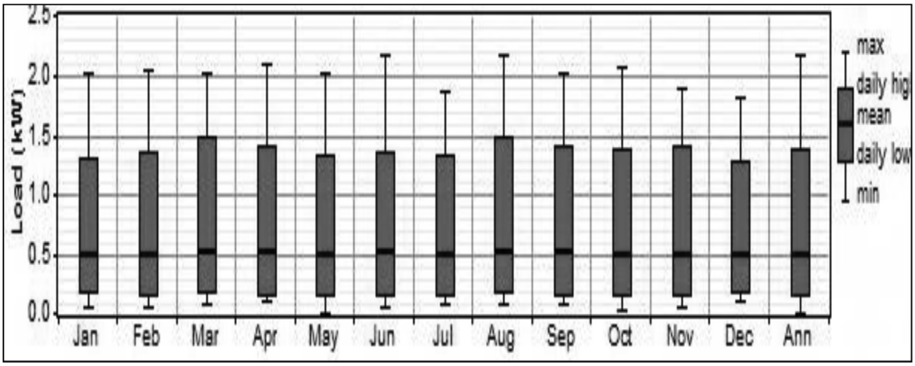


Figure 6: Variations of the monthly household load profiles for the studied community

SYSTEM SPECIFICATION

In this study, all calculations, simulation and optimisation of the hybrid PV-BD energy system and economic performance parameters were done by the HOMER software for a proposed project of a hybrid energy system for households in Omavovwe community. The HOMER software design specifications, in this study, are given Table 2. The PV module, diesel generator and battery type where chosen because of their technical and availability in the Nigeria market. Specifically, the battery, Trojan T-105, has a high market rating and is readily available in the HOMER software data base.

Table 2.
Techno-economic specification for the hybrid PV-BD energy system

S/No	Description	Specification
1	*PV modules (DB F80)*	Polycrystalline [14]
	Nominal power	80 W
	Nominal load voltage	17.5 V
	Short circuit current	5.03 A
	Bus voltage	24 V
	Nominal efficiency	12 %
	Nominal operating temperature	47 ºC
	Temperature coeff. of power	0.97 %/ºC
	Capital cost	300 US$
	Replacement cost	300 US$
	Operating and maintenance cost	10 US$/year [12]
	Life time	20 yrs
	Slope	8.8125º [9]
	PV module floor area	0.42 m^2
2.	*Diesel Generator*	Caterpillar
	Cost/kW	600 US$/kW
	Replacement	600 US$/kW
	Operation and Maintenance	0.15 US$/hr
	Fuel cost	1.00 – 1.5 US$/Litre
	Minimum load factor	40 %
3.	*Battery bank (Trojan T-105)*	
	Capital cost	90 US$
	Replacement	90 US$
	Operation and maintenance	2 US$/yr
	Nominal voltage	6 V
	Nominal capacity	225 Ah
	Nominal energy capacity of each battery	1.35 kWh
	Minimum battery life	5 yr
	Minimum state of charge	30% [13]
	Life time throughput	845 kWh
4.	*Converter*	
	Capital cost	1000 US$/kW
	Replacement	1000 US$/kW
	Operation and maintenance	100 US$/yr
	Efficiency	98 %
	Life time	20 yrs
	Rectifier capacity relative to inverter	100 %
	Rectifier efficiency	95 %

RESULTS AND DISCUSSION

In this study, the optimal configuration of a hybrid PV-BD energy system was undertaken using NREL's HOMER software computer model, interfaced with in-house developed software, which simplified the task of designing renewable electricity generation systems for either—on-grid or stand-alone applications. HOMER's optimization and sensitivity analysis algorithms allowed the rapid techno-economic evaluation of a large number of technology options whilst accounting for variations in technology costs and energy resource availability. The user interface is convenient and powerful [13]. In the current analysis, the information to be provided includes: electrical loads (Figure 5), renewable resources (such as a typical yearly solar insolation data), component technical details/costs, constraints, controls, type of dispatch strategy, etc. HOMER uses life-cycle cost in its computational algorithms. Its user interface is powerful and friendly. The sensitivity analyses, which are tied to key design parameters, is done automatically.

Optimal Configuration

The system calculation and simulation assumed a project's lifetime of 25 years and real annual interest rate of 6% (the difference between the prevailing interest rate and inflation rate), which is the prevailing real interest factor in Nigeria. The HOMER software block diagram for the considered system configuration is shown in Figure 7.

A PV capacity of 0.8kW is suggested, at least to cover the average daily base load of the household as shown in Figure 6, for the analyses; whereas a single generating set is suggested denoted as Gen in Figure 7. This hybrid PV-BD had a peak load of 2.2 kW and daily electrical energy of 12kWh, see Figure 7. The base case for the design uses the annual average global radiation of the zone, 4.4 kWh/m^2/day, with an annual average clearness index of 0.442; and diesel fuel price of 1.50 US$/Litre. Results shown in Figure 8 are the displayed top-ranked system configurations, which are listed according to their net present cost (NPC) for possible system configuration type.

According to the results shown in Figure 8, the optimal hybrid PV-BD energy system comprises 0.8kW PV panels, one 1.5kW diesel generator, 1.5kW converter and twenty-five batteries with life span of eight years. The cost of energy (COE) of the optimal hybrid PV-BD energy system is 0.781 US$/kWh and the corresponding initial capital required,

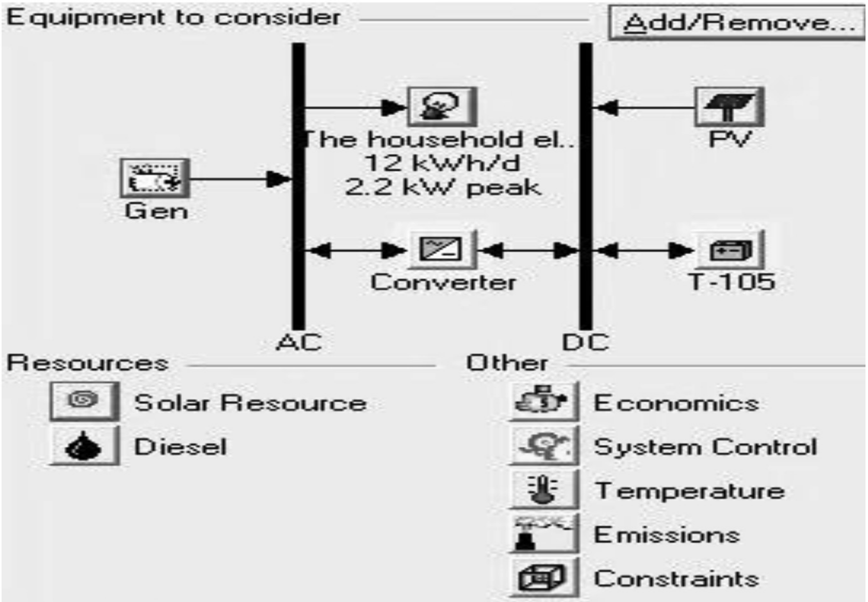


Figure 7: Physical model of the hybrid PV-BD system in HOMER environment

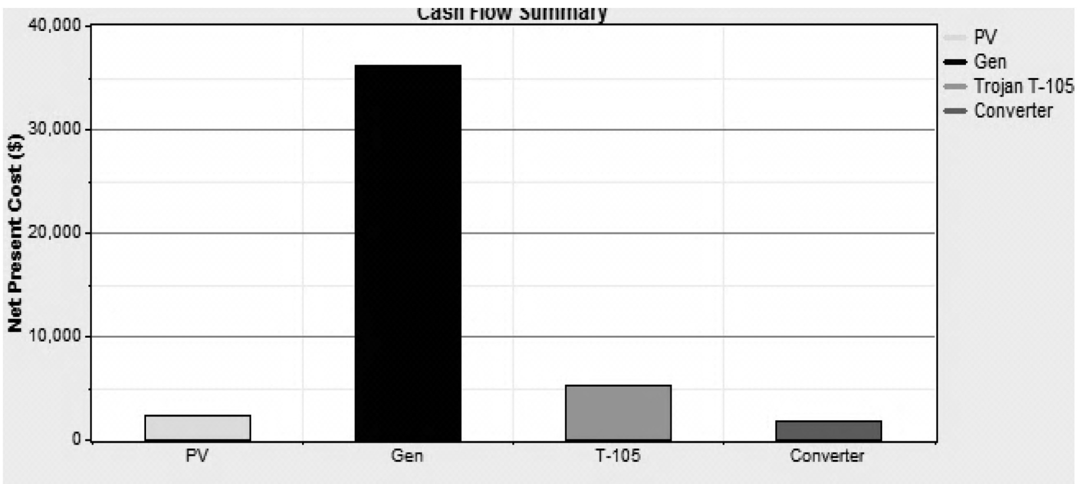
Global Solar (kWh/m²/d) 4.4 Diesel Price ($/L) 1.5 PV Capital Multiplier 1 PV Replacement Multiplier 1

Double click on a system below for simulation results.

	PV (kW)	Gen (kW)	T-105	Conv. (kW)	Disp. Strgy	Initial Capital	Operating Cost ($/yr)	Total NPC	COE ($/kWh)	Ren. Frac.	Diesel (L)	Gen (hrs)	Batt. Lf (yr)
	0.8	1.5	25	1.50	CC	$ 6,900	3,026	$ 45,580	0.781	0.01	1,597	2,999	8.1
	0.8	1.5	30	1.50	CC	$ 7,350	3,024	$ 46,002	0.789	0.01	1,600	3,004	9.8
	0.8	1.5	25	2.30	CC	$ 7,620	3,115	$ 47,445	0.813	0.00	1,624	3,049	7.5
	0.8	1.5	30	2.30	CC	$ 8,070	3,107	$ 47,791	0.819	0.00	1,625	3,051	9.1
		1.5	25	1.50	CC	$ 4,500	3,778	$ 52,794	0.905	0.00	2,008	3,770	7.3
		1.5	30	1.50	CC	$ 4,950	3,769	$ 53,136	0.911	0.00	2,009	3,772	8.8
		1.5	25	2.30	CC	$ 5,220	3,862	$ 54,585	0.936	0.00	2,032	3,816	6.8
		1.5	30	2.30	CC	$ 5,670	3,852	$ 54,911	0.941	0.00	2,035	3,821	8.3

Figure 8: HOMER top-ranked NPC of system configuration.

annual operating cost and NPC are 6,900US$, 3,026US$ and 45,580US$, respectively. The detailed cost distribution of the hybrid PV-BD energy system and summarised NPC of the proposed system components are given in Figure 9.



Component	Capital ($)	Replacement ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
PV	2,400	0	0	0	0	2,400
Gen	900	3,120	1,610	30,630	-105	36,154
Trojan T-105	2,250	2,815	639	0	-486	5,217
Converter	1,350	563	0	0	-105	1,808
System	6,900	6,497	2,249	30,630	-697	45,580

Figure 9: Summarised NPC of the proposed hybrid PV-BD energy system.

The optimal system has 23.5% PV penetration, which falls in the world deployed renewable energy penetration range of 11- 25%. This would be a viable choice for implementation as the penetration of the renewable resource, the PV, is quite significant. The proposed system compared fairly well with a similar system (hybrid PV-BD) in a Malaysian's condition, which gives COE of 0.796US$/kWh for a solar irradiance and cost of fuel of 5.51kWh/m^2/h and 2.03US$/L, respectively [12]; also compared well with a similar system in the Niger-Delta region that gives 0.673US$/kWh for an average solar irradiance and cost of fuel of 3.75 kWh/m^2/day and 1.02US$/L, respectively [7]. A diesel standalone system gives COE of 0.905US$/kWh (see Figure 8) and an annual operating cost of 3,778US$. The COE of the standalone diesel generator justifies the implementation of the hybrid PV-BD. More so, it should be noted that the price of diesel fuel is unstable, always on the increase, in the Nigeria market and there is always artificial scarcity, which hikes the diesel fuel price to an unbearable price. It is envisaged that the price of diesel would soar with the on-going FGN's reforms, which include subsidy removal on petroleum products.

Figure 10 shows average monthly electrical power production of the hybrid PV-BD energy system, which culminates to yearly electrical energy production capacity of 5.571 MWh.

The PV module contributes 19% of the total energy production per year, whereas diesel generator contributes 81% toward the hybrid sys-

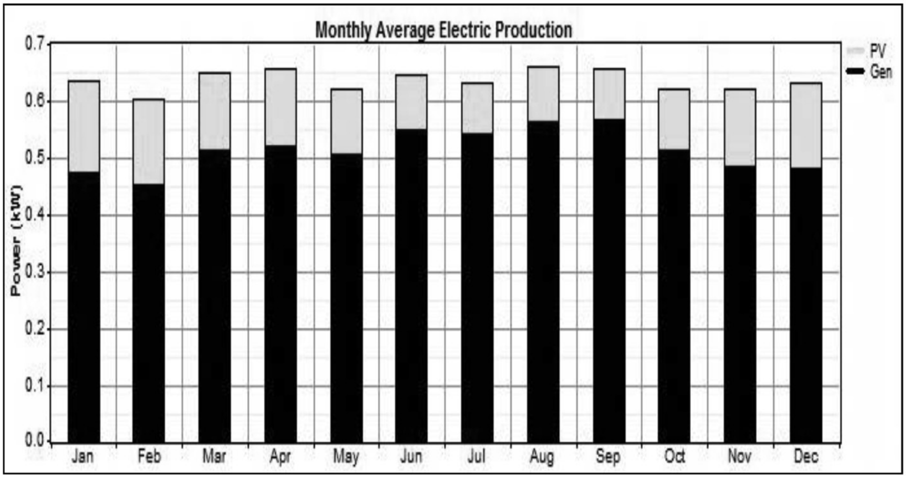


Figure 10: Average monthly electrical power production

tem. The amount of kg of CO_2 emitted per litre of diesel, which is heavily dependent on the generator load and running hours, has a degrading consequence on the environment. The CO_2 emission from a wholly DG energy system is 5,288 Kg/yr; whereas it is 4,206 kg/yr for the optimised hybrid PV-BD energy system, which amounts to 25.72% yearly CO_2 emission reduction. This emission reduction is a significant achievement as it will reduce the over-polluted Niger-Delta region as a result of gas flaring and oil spillage resulting from crude oil production activities [3]. More so, cost imposed on CO_2 emission by environmental legislations, which is the normal practice in most developed nations [15], will be averted. Although, there are currently no such environmental legislations in Nigeria, however, the 25.72% carbon reduction would make the environment friendlier.

With the ongoing FGN's campaigns on the utilisation of renewable energy, of course, which necessitated the establishment of 7.5MW joint venture solar panel manufacturing plant in Nigeria by NASEI, it is envisaged that the prices of solar PV would reduce appreciably. More so, foreseen commitments, as a result of canvassed improvement of oil producing companies' relation toward their host communities, to directly supply diesel fuel to the oil producing communities at a subsidized diesel pump price by the producing companies would have a positive effect on the economy attractiveness of the hybrid PV-BD. Thus, Figure 11 gives the optimal results for the hybrid PV-BD energy system on envisaged 20% reduction of PV cost and 1.00 US$/L diesel price. The off base case has seen the initial capital required, operating cost, NPC and COE of the optimal system being reduced by 3.92%, 35.88%, 30.63% and 30.60%, respectively, compared to the reality case (0.8kW PV, 4.4 kWh/m^2/day, 1.50 US$ litre).

Another possible way of reducing the COE of the base case (0.8kW PV, 4.4 kWh/m^2/day, and diesel fuel price of 1.50 US$ litre) is by increased PV penetration, say 2.5kW, which is now considered as the third case. The economic indices for the third case are compared with the two previous cases as shown in Table 3. If the conditions for the second case (the off base case) are met, it has the best potential for providing the energy needs of households in the community—relatively low initial cost required and COE. Though the third case COE is lowest for all the three cases considered, see Table 3, however, the initial capital cost required is quiet high; of course no household in the oil producing community would be able to afford the initial cost required. Hence, in the light of

Sensitivity variables

Global Solar (kWh/m²/d) 4.4 Diesel Price ($/L) 1 PV Capital Multiplier 0.8 PV Replacement Multiplier 0.8

Double click on a system below for simulation results.

PV (kW)	Gen (kW)	T-105	Conv. (kW)	Disp. Strgy	Initial Capital	Operating Cost ($/yr)	Total NPC	COE ($/kWh)	Ren. Frac.	Diesel (L)	Gen (hrs)	Batt. Lf. (yr)
0.8	1.5	25	1.50	CC	$ 6,420	2,227	$ 34,890	0.598	0.01	1,597	2,999	8.1
0.8	1.5	30	1.50	CC	$ 6,870	2,224	$ 35,297	0.605	0.01	1,600	3,004	9.8
0.8	1.5	25	2.30	CC	$ 7,140	2,303	$ 36,585	0.627	0.00	1,624	3,049	7.5
0.8	1.5	30	2.30	CC	$ 7,590	2,295	$ 36,925	0.633	0.00	1,625	3,051	9.1
	1.5	25	1.50	CC	$ 4,500	2,774	$ 39,959	0.685	0.00	2,008	3,770	7.3
	1.5	30	1.50	CC	$ 4,950	2,765	$ 40,295	0.691	0.00	2,009	3,772	8.8
	1.5	25	2.30	CC	$ 5,220	2,803	$ 41,054	0.704	0.00	2,012	3,777	7.2
	1.5	30	2.30	CC	$ 5,670	2,791	$ 41,345	0.709	0.00	2,011	3,776	8.7

Figure 11: Off base case HOMER top-ranked NPC of the system configuration.

present reality in the oil producing communities, therefore, the base case has the best potential of supplying the electrical energy needs of households in the community as capital costs are reduced by 45.6% when compared to the third case.

Table 3: Comparison of economic indices

Cases		Initial cost required (US$)	NPC (US$)	Operating cost (US$)	COE (US$/ kWh)
1	Base case (0.8kW PV, 4.4 kWh/m^2/day, 1.50 US$ litre)	6,900	45,580	3,026	0.781
2	Off base case (0.8kW PV, 4.4 kWh/m^2/day, 1.00 US$ litre and 20% reduction in PV cost)	6,420	34,890	2,227	0.598
3	Base case (2.5kW PV, 4.4 kWh/m^2/day, 1.50 US$ litre)	12,000	32,894	1,634	0.564

Optimal Design Analyses

Since the optimal configuration of the hybrid PV-BD system is sorted, it is, therefore, easy to obtain optimal sizing of the system by using the in-house software developed as an MS Excel add-in, which has been used in [14], to give data in Table 4.

Table 4: Optimal design parameters

S/No	Quantity	Units	Value
1	Number of PV panel	-	10
2	Total PV floor area	m^2	4.26
3	Number of PV panel in series	-	1
4	String number of panels in parallel	-	10
5	Number of battery in series	-	4
6	Number of battery in parallel	-	2

The rating of the diesel generator is determined by the load. The optimal simulated diesel generator load for the proposed system is 1.5 kW. Hence, we select a 2 kVA generator type (caterpillar) with three phase output AC voltage (3X240 V), 50 Hz with a power factor of 0.85 [16].

Once the sizing of the battery bank is known, one proceeds to the sizing of the voltage regulator. A good voltage regulator, in this case, must be able to withstand the maximum current produced (is 1.5 kW for the optimal system considered) in the system. The input of inverter has to be matched with the battery block voltage while its output should fulfill the specifications of the electric appliances in used, which is specified as (3X240), 50Hz.

CONCLUSION

Adequate electric power production plays an epicentre role in national development; however, there is gross electric power poverty in Nigeria [17]. The situation is worst in the remote villages, which are located far away from the national utility grid. In particular the oil producing communities of the Niger-Delta region, which are mainly located away from towns, are largely in the rugged terrains. These remote communities normally, outside the environmental and health challenges facing them, lack in roads, pipe borne water and, ultimately, supply of electricity, since it is considered impractical to extend the national grid to these dispersed populated areas. Meanwhile, electricity is required for such basic developmental services as pipe borne water, health care, telecommunications and quality education. Therefore, an optimal configuration and design of a hybrid PV-BD energy system for households in Omavovwe community, which has geographical coordinates of 5.32°N, 6.46°E, is done.

The HOMER software computer model is used to determine the most economic system for the hybrid PV-BD energy system with a daily energy load of 12 kWh. The configuration of the optimal hybrid system is selected based on top-ranked system configuration, according to the NPC. The optimal system gives the best components and design with appropriate operating strategy to provide an efficient, reliable and cost-effective system. The optimal system showed that the hybrid PV-BD energy system requires 10 PV panels (0.08kW each), 2 kVA DG, and 25 batteries (225Ah, and 6V each) to produce electricity for a household in the community, which has an average annual irradiance of 4.40 kWh/m^2/day. The COE of the hybrid PV-BD system is 0.781US$/kWh, whereas the initial capital required, annual operating cost and NPC are 6,900 US$, 3,026 US$ and 45,580 US$, respectively. Similar study for the utilization of hybrid PV-BD in the Niger-Delta region considered here gives COE of 0.673US$/kWh for an average solar irradiance and cost of fuel of 3.75 kWh/m^2/day and 1.02US$/L, respectively, [7] as against the 0.781 US$/kWh obtained in the present study. The difference is attributed to the cost of fuel used in the previous study. Also, the hybrid PV-BD system gives CO_2 emission reduction, which amounts to a yearly 25.72% reduction. This emission reduction is a significant achievement as it will reduce cost imposed on CO_2 emission by environmental legislations, which is the normal practise in most developed nations [15]. Although, there are currently no such environmental legislations in Nigeria, but the 25.72% carbon reduction would make the environment friendlier. It is observed that positive drives by the FGN and the oil producing companies for the utilisation of PV energy in the oil producing communities would see the initial capital required, operating cost, NPC and COE of the optimal system being reduced by at least 3.92%, 35.88%, 30.63% and 30.60%, respectively. The proposed hybrid PV-BD system has the potential of transforming the poverty embroiled oil producing communities in the Niger-Delta region of Nigeria to enviable reputes—social and economic.

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